

Crop planning in the Market Garden



THE
MARKET GARDENER'S
MASTERCLASS

5 steps to your planning

1. ESTABLISH YOUR OBJECTIVES AND FINANCIAL GOALS
2. ESTABLISH YOUR GAME PLAN
3. ESTABLISH YOUR OPERATING CALENDARS
4. ESTABLISH THE MAP OF YOUR SEASON
5. ORDER SEEDS, EQUIPMENT AND MATERIAL



STEP1- SET YOUR OBJECTIVES AND FINANCIAL GOALS

- State, determine or reaffirm you mission statement
- Think about and write what goals you want to achieve next season
- Determine what are you financial objective, mainly sales objective



Determine what are you financial objective, mainly sales objective

- A well run market garden with the most of the workload accomplished by the growers can expect to have margins from 35%to 65%
- Sales objectives need to be align with your markets, your experience the size of your operation and your seasonality
- Sales target should be realistic; not conservative, nor pie in the sky



Sales objectives; an example

- A 2,5 acre (1 hectare) market garden with 180 permabeds (100'-30M) long
- 1 greenhouse 42'X110' (13m X33m) and 3 cold houses 30'X100' (9m X30m)
- 3 employees, 1 seasoned grower
- In production 8 months/per year, selling 35 weeks/year
- Good sales outlets and good quality (following the steps of this masterclass!)



- $160\,000\$ \times 50\%$ (profit margins) =
80 000\$ revenue
- Is that enough for your household income ?
- Do you need to work another job
- Can you improve your cost of production ?
- Can you add another profit stream for same number of clients ? Example eggs



160 000K / 35 weeks = 4 500K /week

- Can you handle the workload ?
- Can your market support such weekly sales ?
- Is that seasonality realistic ?
- 4 500K/ years= weeks at double that ? Can you handle 9000K? Can you delivery truck support it ?



STEP 2- ESTABLISH YOUR GAME PLAN

- Determine your sales outlets
- Determine what your growing for whom and it what quantity
- Your on your own with your game plan



Game plan: example

- **Selling to chef on Wednesday**
- **Selling to farmers market on saturday**
- **Have a small CSA pick-up on Thursday night**
- Could be anything....grocery stores, buyers coop, fat CSA etc...



What crop are you growing?

- Are you growing everything? really?
- Inexperienced growers should perhaps limit themselves to 30 crops
- Focus on high value (yields/space/time) crop
- Focus on crops that mechanize farm can harder handle (ex mesclun, roots with their tops on, greens etc)



LIVES		GAME PLAN		CROP CHART		SALES FORECAST		CROP MAP		CROP MAP FV		DIRECT SEEDING CALENDAR									
A		B		C		D		E		F		G		H		I		J		K	
 THE MARKET GARDENER'S MASTERCLASS CROP CHART																					
Crop		rows /30'' beds	Spacing on the	DTM	Harvest window	Total days in garden	Days in cell	Yields / 100' bed	Price/unit	Unit	Revenue per bed										
Nantaises Carrots		6	2''	60	14	81	X	130	3,00 \$	Bunch	390,00 \$										
Salanova Lettuce		4	6					130	10,00 \$	\$/livre (moyenne 1 livre /	1300,00 \$										
Dill		5	5	45				600	2,00 \$	tête	1200,00 \$										
Basil		4	12	70				400	3,00 \$	tige	1200,00 \$										
Coriander		5	5	60	14			600	2,00 \$	unité	1200,00 \$										
Mini Fennel		4	6	60				800	1,50 \$		1200,00 \$										
Greenhouse English Cucumber		1	18	55				600	2,00 \$	unité	1200,00 \$										
Green Onion			6	65				500	2,00 \$	botte (note: botte de 4 à	1000,00 \$										
Brussel Sprouts		2	18	110	21			130	7,00 \$	tige	910,00 \$										
Swiss Chard		2	12	60	84			360	2,50 \$	botte	900,00 \$										
Kale		2	18	55				360	2,50 \$	unité	900,00 \$										
Lettuce		3	12	48				300	3,00 \$	unité	900,00 \$										
Rutabaga		3	8					450	1,50 \$	110g / planche	900,00 \$										
Pak Choi		4	6	45				800	1,00 \$	unité	800,00 \$										
Leek		3	6	75-100				600	1,25 \$		750,00 \$										
Winter Radish		3	8					450	3,00 \$		675,00 \$										
Rapini		3	8	45				450	3,00 \$	\$/livre (moyenne .5 livre /	675,00 \$										
Garlic		3	6	obre-mi-juillet				600	1,00 \$	unité	600,00 \$										
Summer Squash		1	24	50	42			600	1,00 \$	unité	600,00 \$										
Kohlrabi		4	12	45	14			400	1,50 \$	unité	600,00 \$										
Bell Pepper		1	10.5	80				140	4,50 \$	lbs	630,00 \$										
Cauliflower / Romanesco		2	18	70	14			130	4,00 \$	unité	520,00 \$										
Aubergine / Eggplant (tunnel)		1	12	mai-octob	72			125	4,00 \$	lbs	500,00 \$										
Beetroot TR		3	4	60	14			130	3,00 \$	botte	390,00 \$										
Beetroot SD		3	g : LJ12 : 13-	60	14			130	3,00 \$	botte	390,00 \$										
Nantaises Carrots		6	ng : X-24: 11-	60	14			130	3,00 \$	botte	390,00 \$										
Celery Root		3	12	140	14			300	1,50 \$	unité	450,00 \$										
Hakurei Turnip		6						150	3,00 \$	bottes	450,00 \$										

CROP	RESTURANT	MARKET	CCSA
FRESH CARROTS	X	X	X
TOMAOTOS	X	X	X
CHERRY TOMAOTES	X	X	X
MESCLUN MIX	X	X	X
ARRUGULA	X	X	X
ASIAN GREENS	X	X	X
HAKUREI TURNIPS	X	X	X
SMALL RADDISHS	X	X	X
FRESH BEETS	X	X	X
PEPPERS	X	X	X
EGGPLANTS	X	X	X
GARLIC		X	X
GREEN ONIONS		X	X
SUMMER SQUASH		X	X
SPINACH		X	X
ENGLISH CUCUMBER		X	X
LETTUCE		X	X
LIBANESE CUCUMBERS		X	X
ONIONS		X	X
SWISS CHARD		X	X
KALE		X	X
ONION		X	X
BROCOLI			X
KOHLRABI			X
PEAS			X
BEANS			X
ETC, ETC.....			X

Now determine how much you need off each

- Estimate how much your going to sell of each product / sales outlets
- Calculate your production in terms of yield/ beds
- The bed is your unit of measure



And then, from when to when are you growing it

- Chefs like reliability and constance
- The first one a market is the king
- Csa member will appreciate balance shares
- I could go on and on....



Could look like this....

Carrots

Harvest the equivalent of 2 beds per week from April to December. Estimated 250 bunch/week

- **16 beds** of storage carrots for our winter carrots. Best date for this is around august 5
- **4 beds** Early seedlings will be planted in tomato house early January and should be should 8 weeks later.
- **4 beds** will go in the cold house mid-february 4 beds
- **4 more** beds will be planted mid-mars in caterpillar tunnels
- **2 beds** planted per week from April 15 (first outdoor seedling) to august 8 (triple the last planting)

Notes:

No rainbow carrots

We almost always sell all of our carrots - might plan to grow 3 beds a week, if we have the space

MESCLUN

Harvest from 8 beds of Salanova + 2 beds of frisée from April to December. Estimated 200lbs/week

10 beds every 3 week with multiple harvest.

Transplanted from mars to 1 to October 25

First indoor planting is XXXX

Last indoor planting is XXXX

SO ON AND SO FORTH

ESTABLISH YOUR GAME PLAN...

CONTINUED

- Make sure your on target with your sales objectives
- Make sure you have the space in the garden
- Crop planning 2.0 would be to measure if you have enough man power....





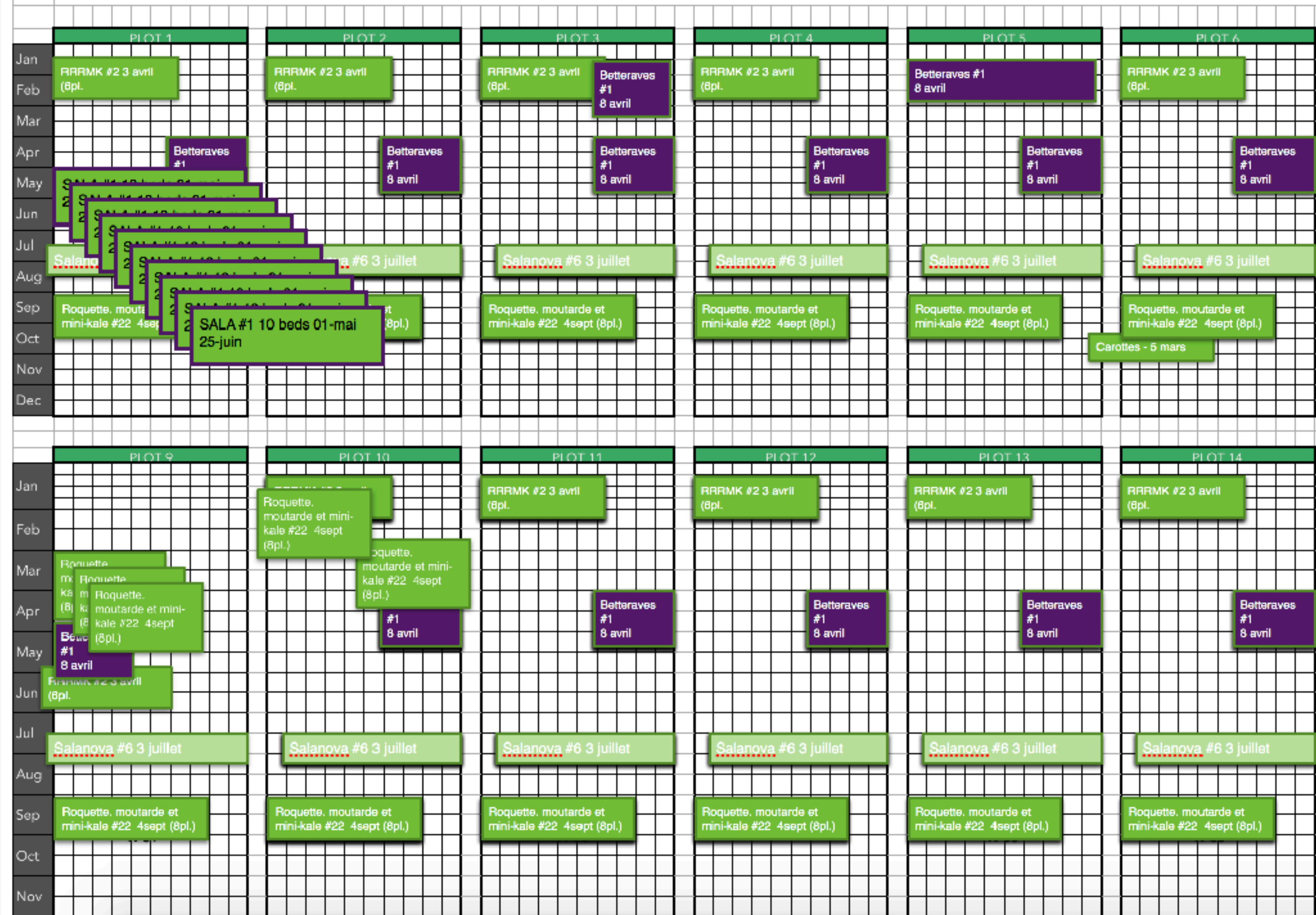
BUDGET PLANNER

Revenu generated

\$ 207 480

4	CARR	Carrots	2	16	32	130	Bunch	\$ 3,00	\$ 390	\$ 12 480,00
5	SALA	Salanova	10	15	150	130	lbs	\$ 10,00	\$ 1 300	\$ 195 000,00
6	RRRMK	Radishes								\$ -
7	RRRMK	Hakurei Turnips								\$ -
8	RRRMK	Mini Kale								\$ -
9	RRRMK	Mustard								
10	RRRMK	Arugula								\$ -
11	Broccoli & Cabbage	Broccoli								
12	Broccoli & Cabbage	Chinese Cabbage			0					\$ -
13	Broccoli & Cabbage	Savoy Cabbage			0					\$ -
14	Broccoli & Cabbage	Red Cabbage (Ruby)			0					\$ -
15	Broccoli & Cabbage	Fall Green Cabbage			0					\$ -
16	Broccoli & Cabbage	Pointed Green Cabbage (Caraflex)			0					\$ -
17	Broccoli & Cabbage	Summer Green Cabbage (Farao)			0					\$ -
18	Focak	Dill								\$ -
19	Focak	Kohlrabi								
20	Focak	Cilantro								\$ -

CROP MAP



STEP 3 -ESTABLISH YOUR OPERATING CALENDARS

- Pre-planning is winning strategy
- Systemize all your complexity into simple procedures to follow
- IMPOSSIBLE to remember everything, so don't try!
- Dynamic calendars are your best friends





EST. HARVEST TIME

TRANSPLANT CALENDAR

Seeding date	REEL seedin g date	# of Bed s	spaci ng on the row (inch es)	# of Trays /100' Bed	Total # of Trays (includes a security of 30%)	Tray Size	# of Days in Tray	Co mpl ete d	By Wh om	Notes	EXP.Trans pl. Date	REEL. Transpl. Date	DTM	Days of Harves t	Plot	# of Bed s	#r o w s p e r b
9/2	9/2	5	4	6	8,125	40,6	128	30			11/3	11/3	55	21	SJ	5	4
28/2	28/2	12	4	6	8,125	97,5	128	30			30/3	30/3	55	21	TF	12	4
19/3	19/3	8	4	6	8,125	65,0	128	30			18/4	18/4	55	21	TC	8	4
9/4	9/4	10	4	6	8,125	81,3	128	30			9/5	9/5	55	21	F	10	4
30/4	30/4	10	4	6	8,125	81,3	128	30			30/5	30/5	55	21	F	10	4
21/5	22/5	10	4	6	8,125	81,3	128	30			20/6	21/6	55	21	F	10	4
11/6	11/6	10	4	6	8,125	81,3	128	30			11/7	11/7	55	21	F	10	4
2/7	2/7	10	4	6	8,125	81,3	128	30			1/8	1/8	55	21	F	10	4
23/7	23/7	10	4	6	8,125	81,3	128	30			22/8	22/8	55	21	F	10	4
13/8	13/8	20	4	6	8,125	162,5	128	30			12/9	12/9	55	21	TC	20	4
15/9	15/9	18	4	6	8,125	146,3	128	30			15/10	15/10	55	21	TF	18	4
9/2	9/2	?	4	6	8,125	▲	128	30			11/3	11/3	42	21	SJ	?	4
28/2	28/2	?	4	6	8,125	▲	128	30			30/3	30/3	42	21	TF	?	4
19/3	19/3	2	4	6	8,125	16,3	128	30			18/4	18/4	42	21	F	2	4
9/4	9/4	2	4	6	8,125	16,3	128	30			9/5	9/5	42	21	F	2	4
30/4	30/4	2	4	6	8,125	16,3	128	30			30/5	30/5	42	21	F	2	4
21/5	21/5	2	4	6	8,125	16,3	128	30			20/6	20/6	42	21	F	2	4
11/6	11/6	2	4	6	8,125	16,3	128	30			11/7	11/7	42	21	F	2	4
2/7	2/7	2	4	6	8,125	16,3	128	30			1/8	1/8	42	21	F	2	4
23/7	23/7	2	4	6	8,125	16,3	128	30			22/8	22/8	42	21	F	2	4
13/8	13/8	2	4	6	8,125	16,3	128	30			12/9	12/9	42	21	TC	2	4
15/9	15/9	2	4	6	8,125	16,3	128	30			15/10	15/10	42	21	TF	2	4
5/3	5/3	1	4	6	8,125	8,1	128	30			4/4	4/4	45	14	TF	1	4
19/3	19/3	1	4	6	8,125	8,1	128	30			18/4	18/4	45	14	F	1	4
2/4	2/4	1	4	6	8,125	8,1	128	30			2/5	2/5	45	14	F	1	4

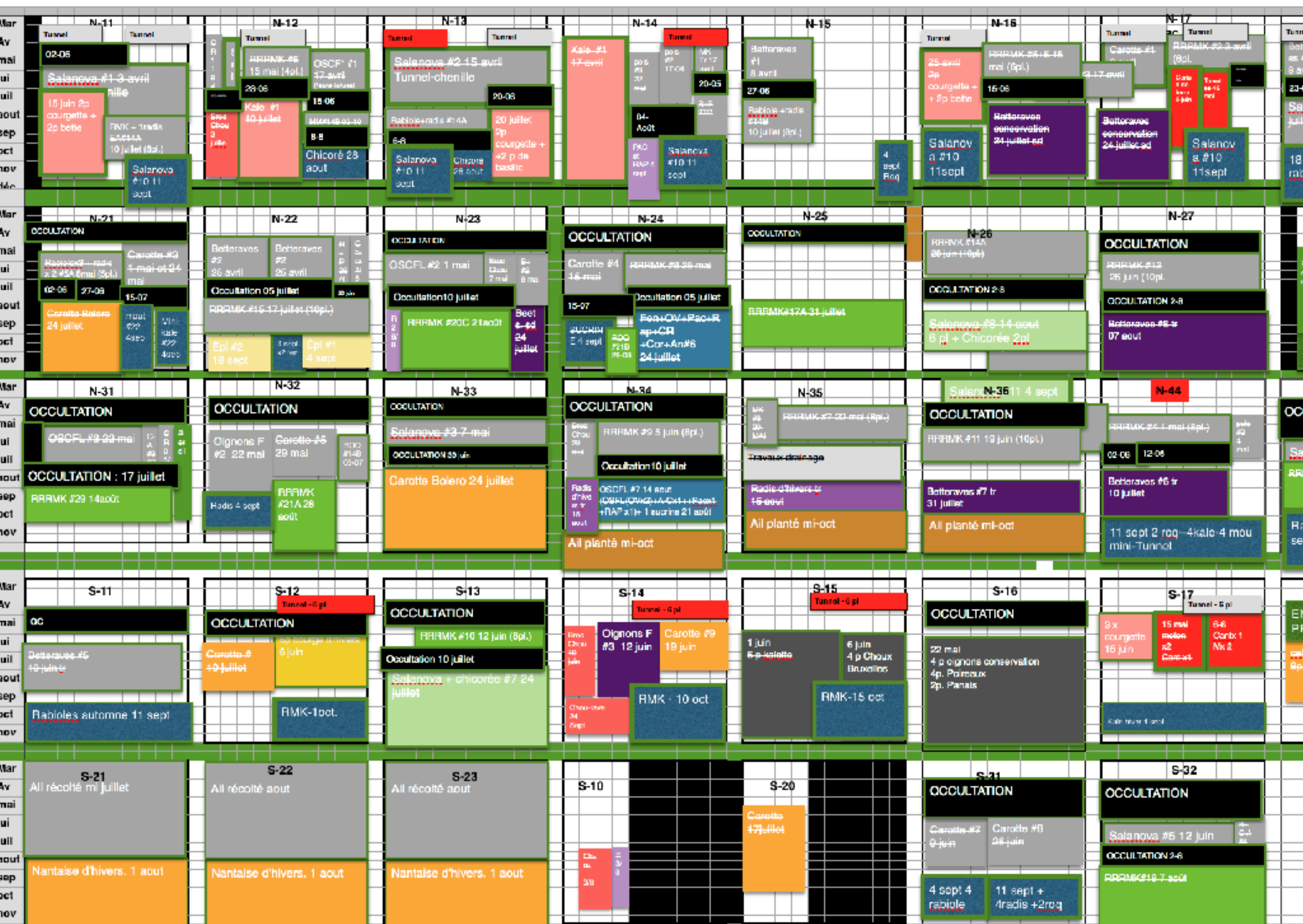
**Task, number of days after transplanting / seeding**

CODE	CROP	At seeding/ transplanting	Stale seed bed	flame weeder	flexline harrow-1	flexline harrow-2	Biodisc	Hoe	handweed	Boron/ seaweed -1	Boron/ seaweed -2	Straw mulch+topdr oss	head pinch/ chop	mow and tarp
RRRMK	Radish	Net/row cover			10	17								
	Turnip													
	Arugula													
	Mini Kale													
	Mustard													
CARR	Carrots		-7	5			21		26					
SALANOVA	Salanova	Landscape fabric												
	Frisee													
BEET	Beets						12	19		20				
BEET	Storage Beets						20	27		30				
FOKAC	Fennel				10		17							
	Cilantro							17						
	Dill							17						
	Green Onion						17							
	Kohlrabi							17						
SPARL	Lettuce	Net/row cover						12						
	Rapini													
	Bok Choi													
	Baby Romaine													
BASIL	Basil												21	21
SSQ	Summer Squash													
GARL	Garlic													
WRAD	Winter Radish													
CELK	Mini Celery Root									20	34			
CA-BRO	Cabbage-									24		24		

STEP 4- ESTABLISH THE MAP OF YOUR SEASON

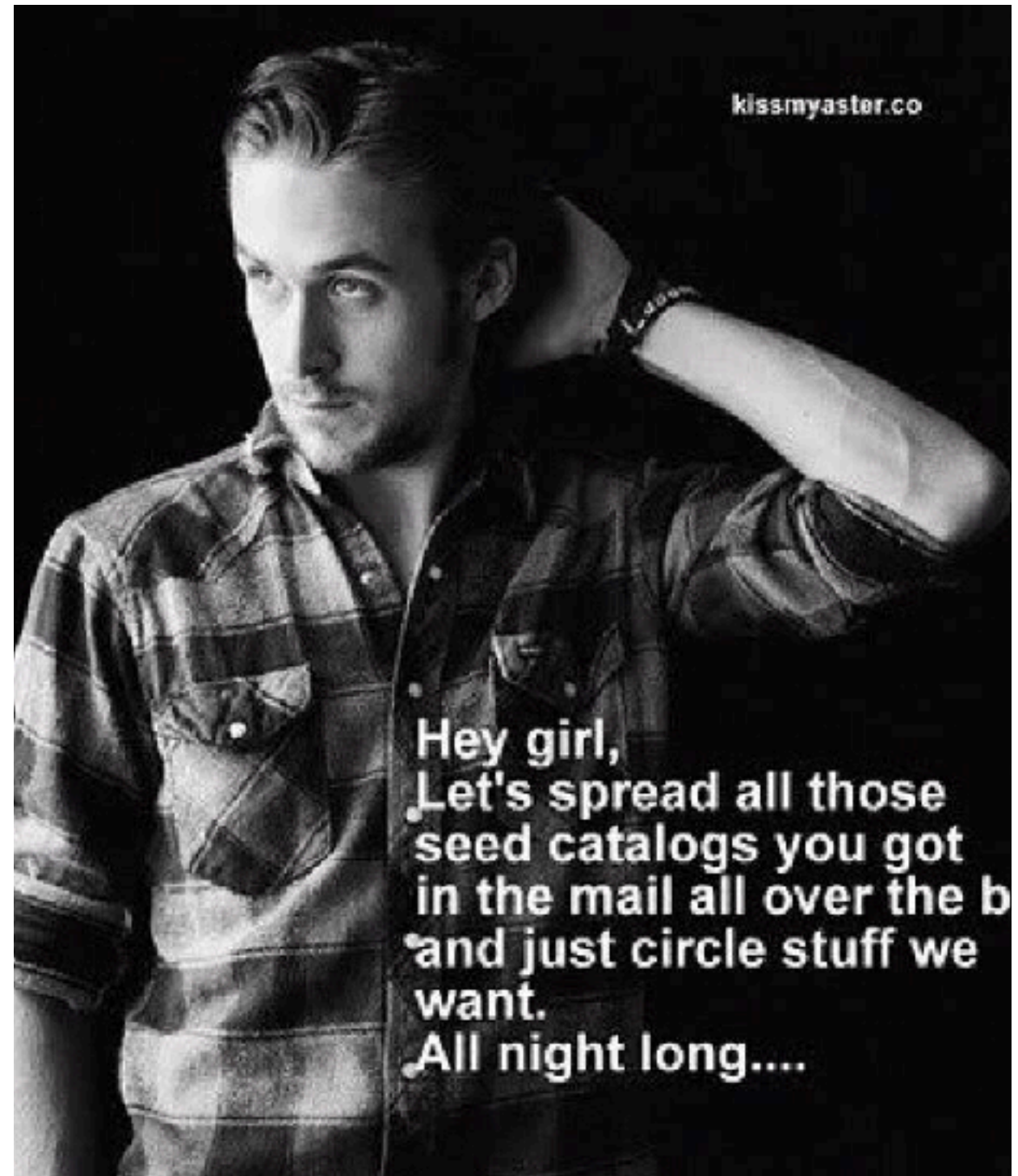
- Make sure everything fits
- Allows to manage secessions in a timely manner
- Think about where you puts crops with regard to one another. We've group hours
- Think about irrigation, insect nets, row covers, etc. **EVERYTHING** that will help you streamline the management of your crop





STEP 5-ORDER SEEDS, EQUIPMENT AND MATERIAL

- Choose cultivars
- Order seeds for the whole season
- Start with fresh seeds if possible
- Shop around \$\$\$





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Species	Cultivar	Provider	Photo	DTM	Information from the provider
Carrots	Napoli	JSS		58	Ideal for overwintering. Organic seed with NOP-compliant pelleting. A specialized variety with a sweet taste when sown in fall for winter outdoors in the open or protected with fabric row cover. Up to 7" c
Carrots	Bolero	JSS			Ideal storage carrot. Conventional seed with NOP-compliant pelleting. Medium-long, 7-8", roots are uniform, thick, slightly tapered, and bl fresh, and tops the charts after long-term storage. A great choice for blight and powdery mildew; and intermediate resistance to cavity sp
Garlic	Music	Ferme Tournesol			The standard Canadian hardneck garlic. Hardneck. 4-6 huge cloves/b
Garlic	Quebec	Ferme Tournesol			Our earliest maturing Rocambole. Hardneck. 6-10 Large easy to pee but later than Artichokes. Stores until Jan-Feb.
Garlic	Siberian	Ferme Tournesol			This is a stunning garlic. Purple marbling on a white background. In mature in our fields. From Siberia. Hardneck with 5-7 large hot & pu
Dill	Bouquet	HMS		40-50 leaf	The most widely-grown organic dill for fresh eating, flowers and seed seedhead production and cut flowers. Produces seed heads well ahead average 6" wide. 15M seeds/oz. Great for pickling - 30" tall (Anther
Aubergine / Eggplant	Angela F1	JSS		65	High-yielding striped Italian type, strong plants for the greenhouse.



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SEED ORDERING

Vegetable	Seed /unit	Cultivar	#seedi ngs	beds/ seedings	rows/ bed	Spacing on row	Security	#seed	Column2
Carrot fresh	1	Napoli	16	2	6	1,5	1,3	199680	
Carrot winter	1	Bolero	1	16	6	1,5	1,3	99840	
								▲	
								▲	
Salanova	0,2	20% green incised	14	8	4	6	1,15	20608	
Salanova	0,3	30% green sweet crips	14	8	4	6	1,15	30912	
Salanova	0,3	30%Red sweet crisp	14	8	4	6	1,15	30912	
Salanova	0,2	20%Red butter	14	8	4	6	1,15	20608	
Frisee	1	Rhodos OG JSS	13	2	4	6	1,3	27040	
Frisee	1	MTF - test WGS	2	1	4	6	1,3	2080	
Little Gem Mini Lettuce	1	Bambi JSS	10	0,5	4	6	1,3	5200	
Little Gem Mini Lettuce	1	Dragocn JSS	10	0,5	4	6	1,3	5200	
Little Gem Mini Lettuce	1	Winter density HMS	2	1	4	6	1,3	2080	
Pac Choy	1	Shanghai green HMS	12	0,5	4	6	1,3	6240,00	10000seed pe
	1	Tatsoi	12	0,5	4	6	1,3	6240,00	
Rapini	1	Zamboni WDS	6	1	3	8	1,3	3510,00	
Rapini	1	Sorento WDS	6	1	3	8	1,3	3510,00	
Lettuce	0,20	Eruption URS	12	1	3	12	1,3	936	
	0,20	Black seeded simpson H	12	1	3	12	1,3	936,00	
	0,20	Reine de glace	12	1	3	12	1,3	936,00	
	0,20	Pirat	12	1	3	12	1,3	936,00	
	0,20	New red fire	12	1	3	12	1,3	936,00	
	0,20	Sylvesta JSS	12	1	3	12	1,3	936,00	
Fresh Onion	4	Purplette	4	6	3	8	1,5	64800	
		Sierra					1,5	▲	
		Goldcoin					1,5	▲	
							1,5	▲	
Basil	1	aroma	5	1	4	12	1,3	2600	

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